

Samy Badreddine

RESEARCH SCIENTIST · PHD

📞 (+34) 651453775 | ✉ badreddine.samy@gmail.com | 🏠 <https://sbadredd.github.io> | 📷 sbadredd | 🌐 sbadredd | 📄 Full Publications

Summary

Machine learning researcher and engineer with a background in neurosymbolic methods, probabilistic inference, and applied mathematics. I work on making probabilistic reasoning scalable and expressive with applications in scientific discovery and knowledge graphs. I particularly enjoy turning these ideas into easy-to-use software libraries.

Education

University of Trento

Hybrid

PHD IN NEUROSYMBOLIC AI AND PROBABILISTIC MACHINE LEARNING

Nov. 2023 - Oct. 2026

- Thesis: *Unifying and Scaling Neurosymbolic AI*, (i) unified language/library for cross-framework training and evaluation, (ii) scalable probabilistic reasoning via knowledge compilation and approximation, (iii) expressiveness analysis on large-scale tasks.
- Supervisors: Luciano Serafini (FBK), Emile van Krieken (VU Amsterdam), Michael Spranger (Sony AI)

University of Edinburgh

Edinburgh, UK

VISITING RESEARCHER

Summer 2025

- Host: Antonio Vergari. Approximate knowledge compilation for weighted model counting (weighted #SAT) in probabilistic neurosymbolic systems, via circuits and tensor networks.

Ecole polytechnique de Bruxelles (ULB)

Brussels, Belgium

MASTER IN COMPUTER SCIENCE AND ENGINEERING (AI SPECIALIZATION)

Sep. 2017 - Jun. 2019

Ecole polytechnique de Bruxelles (ULB)

Brussels, Belgium

BACHELOR IN CIVIL ENGINEERING

Sep. 2014 - Jun. 2017

Research in Industry

Sony AI

Barcelona

RESEARCH SCIENTIST

Oct. 2023 - Present

- Lead research on scaling **probabilistic reasoning** and knowledge compilation with approximation guarantees, with applications to **biomedical knowledge graphs** (complex query answering).
- Improve the expressivity and performance of embedding-based **link predictors** at large scale (ICML 2026).
- Design an open-source unified language and **library** (ULLER) standardizing training and evaluation across **neurosymbolic** frameworks (oral + tutorial, NeSy 2024).
- Mentor three research interns whose projects produced a peer-reviewed publication and features shipped to product applications.

Sony AI

Tokyo

AI ENGINEER

May 2020 - Sep. 2023

- Led research on **Logic Tensor Networks**, which relax constraints to **differentiable logics** used as inductive bias in learning.
- Co-led a project on **biomedical hypothesis generation** via **link prediction over temporal knowledge graphs** (US patent filed).
- Delivered a recommender system end-to-end from research prototype to a deployed app tested by professional users (chefs).
- Mentored three research interns whose projects produced three peer-reviewed publications.

Sony Computer Science Laboratories

Tokyo

MACHINE LEARNING RESEARCH INTERN

Summer 2018 and 2019

- Conducted research in Transfer Learning in **Reinforcement Learning** (RL) and Semantic Image Interpretation with a focus on **neurosymbolic** AI under the mentorship of Dr. Michael Spranger.

Selected Publications

JOURNALS

- | | |
|------|---|
| 2022 | Logic Tensor Networks – AIJ. S. Badreddine, A.A. Garcez, L. Serafini, M. Spranger. AIJ 2026 Prominent Award , Featured in press (<i>Nature news</i>). NeSy Relax Fwk |
| 2023 | Link prediction for hypothesis generation – AI Review. U. Akujuobi, P. Kumari, J. Choi, S. Badreddine, K. Maruyama, S. K. Palaniappan, T. R. Besold. KG AI4S |

CONFERENCES AND WORKSHOPS

- 2026 **On the Theoretical Limitations of Embedding-based Link Prediction – ICML.** S. Badreddine, E. van Krieken, L. Serafini. [KG](#) [LT](#)
- 2026 **Where to Approximate in Neurosymbolic Inference? – CoLoRAI @ ICML.** S. Badreddine, E. van Krieken, L. Serafini, A. Vergari. Workshop oral. Under review at a conference. [NeSy](#) [TN](#) [Prob](#)
- 2025 **Rewarding explainability in drug repurposing with knowledge graphs – IJCAI.** S. Nunes, S. Badreddine, C. Pesquita. [KG](#) [RL](#) [AI4S](#)
- 2024 **ULLER: A Unified Language for Learning and Reasoning – NeSy.** E. van Krieken*, S. Badreddine*, R. Manhaeve, E. Giunchiglia. Oral + tutorial. [NeSy](#) [Fwk](#)
- 2023 **What's Wrong with Gradient-based Complex Query Answering? – NeSy.** O. El Harzli, S. Badreddine, T. R. Besold. Oral – **Best Paper Award.** [KG](#) [NeSy](#) [Relax](#)
- 2019 **Injecting Prior Knowledge for Transfer Learning into Reinforcement Learning using Logic Tensor Networks – NeSy @ IJCAI.** S. Badreddine, M. Spranger. Oral. [NeSy](#) [Relax](#) [RL](#) [Game](#)

PATENTS

- 2025 **Method, apparatus, and computer-readable medium for generating a hypothesis from a knowledge graph – US Patent.** T. R. Besold, U. Akujuobi, S. Badreddine, J. Choi, K. Nishikawa, et al. Application 18/664,454. [KG](#) [AI4S](#)

Topics: [NeSy](#) neurosymbolic [Relax](#) differentiable relaxation [TN](#) tensor networks [Prob](#) probabilistic modeling [KG](#) knowledge graphs [RL](#) reinforcement learning [LT](#) learning theory

Applications: [AI4S](#) AI for science [Fwk](#) framework/library [Game](#) video games

Skills

Programming Languages Python (PyTorch, TensorFlow; einsum/einops), Modern C++, C
French (native), English (full professional), Japanese & Dutch (limited working)